

Maria Loureiro

Machine Learning Engineer · AI for Drug Discovery & Life Sciences

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I build generative AI systems for biotech and pharma companies, and I spend the rest of my time getting the people who have led that industry for decades talking to the generation about to inherit it. Five years across a biotech startup, an AI consultancy and an executive network.

SELECTED HIGHLIGHTS

- **Davos 2026.** 1 of 40 Global Shapers selected worldwide, and the only Portuguese delegate, to the World Economic Forum Annual Meeting.
- **Generative AI for drug discovery.** Agentic AI pipelines on AWS for five-plus US biotechs and a biopharma with ongoing clinical trials.
- **Executive convening in pharma.** Pharma Bridge Day (200+ participants) and three innovation programmes for leadership across ten of the largest pharma companies.

PROFESSIONAL EXPERIENCE

Jul 2024
– Present

Advanced Machine Learning Engineer, Life Sciences

[Loka](#) · Portugal (Remote) · *promoted from Machine Learning Engineer*

- Design and ship generative AI and agentic pipelines on AWS for biotech and pharma clients, including screening candidate molecules for binding affinity against target proteins to make laboratory studies more efficient.
- Work across five-plus US biotechs and one biopharma with ongoing clinical trials, cutting the time and cost of identifying viable drug targets and molecules.

Sep 2024
– Present

Senior Business Manager

[GenH](#) · Portugal (Remote) · *concurrent with Loka*

- Run Pharma Bridge Day, connecting Portuguese pharma professionals with peers in Boston, Basel and beyond: 50+ in the room in Lisbon, 200+ participants in total.
- Deliver three executive innovation programmes of roughly fifteen participants each, for C-suite and country-level leadership drawn from around ten of the largest pharmaceutical companies. Manage one direct report.
- Manage a 100-member alumni network of Portuguese professionals across Europe and the Americas, running initiatives that keep senior leaders and early-career members in contact.

Jul 2021
– Jun 2024

Biomedical Data Scientist

[iLoF](#) · Porto, Portugal · *Acting Data Science Coordinator (Apr–Jun 2024)* · *Biomedical Data Scientist (2022–24)* · *Data Scientist Intern (2021–22)*

- Designed and shipped an internal application giving the whole company 24/7 access to data generated continuously by 24-hour experiments, adopted by 15–20 people as the standard way to view and analyse results.
- As acting coordinator, planned and managed data science work company-wide within a sprint framework, working between the development and product design teams to surface and resolve risks.
- Developed an end-to-end deep learning pipeline for encoding and classifying time-series optical data, including techniques to mitigate the influence of confounding clinical factors.

2019
– 2021

Earlier research & data science roles

[CerCo](#) – Centre de Recherche Cerveau et Cognition, Toulouse · [NILG.AI](#) · [CINTESIS](#), Porto

- Automated longitudinal marmoset brain segmentation at CerCo, replacing a manual process with a standardised pipeline. Built a live AI content-retrieval demo at NILG.AI. Ran a systematic review and signal-processing analysis of fetal heart rate traces at CINTESIS, published in *Frontiers in Pediatrics*.

EDUCATION

2017 – 2022

MSc & BSc in Bioengineering (Biomedical Engineering)

Faculty of Engineering, University of Porto (FEUP) · Grade A

Thesis: Transfer Learning Techniques for Classification of Biological Samples using Optical Fingerprint · Erasmus+ research placement at CerCo, Toulouse (2021)

LEADERSHIP & COMMUNITY

Nov 2024 – Present	Global Shaper Global Shapers Community (World Economic Forum) · Porto Hub - Selected by application as 1 of 40 Global Shapers worldwide, and the only Portuguese delegate, to the WEF Annual Meeting 2026 in Davos. - Lead the Digital Upskilling project in Matosinhos, teaching technology, automation and AI to high school students in a public school. Around 50 students reached so far. - Organised the 2025 Iberian retreat, bringing 80 Shapers from Spain and Portugal together with invited guests in Porto.
Apr 2025 – Present	Volunteer Speaker, Inspiring Girls Portugal — Speak in schools about my own path, to show teenagers that career direction is not decided by gender.
Oct 2025 – Feb 2026	Mentor, Geek Girls Portugal — Mentor women in technology on navigating their careers and developing their skills.
2023	Organiser, MIT Critical Data Health Datathon — Two-day event bringing teams together to build data-driven solutions to healthcare problems.
2018 – 2022	President, NEB FEUP/ICBAS Bioengineering Students' Association — Led a 45-person organisation (2021/22); previously Vice President of the Supervisory Board and Biomedical Engineering representative.

AWARDS & HONOURS

2026	World Economic Forum Annual Meeting Honoree · Global Shapers Community One of 40 Global Shapers selected worldwide, and the only Portuguese delegate, in Davos.
2025	Winner, GenAI Bootcamp Hackathon · Loka with AWS An AI genomics solution automating variant analysis under responsible-AI practices, cutting analysis from six hours to one hour per case.
2021	1st Place, Eurekaathon · NOS A 50-hour data-driven competition on sustainable urban mobility in Matosinhos.
2021	2nd Place, Age-i-Tank Developed a healthcare venture and pitched the investment deck to a venture capital jury.

SELECTED PUBLICATIONS

Feb 2026	<u>A Loka Engineer Goes to Davos</u> — Loka
Nov 2025	<u>Tecnológicas vs Farmacêuticas: onde estão os melhores líderes?</u> — Netfarma
Apr 2025	<u>Será o salário o principal culpado pela dificuldade das empresas em reter talento?</u> — Público
Feb 2025	<u>Showdown in AI Drug Discovery: LLMs vs Diffusion Models</u> — Loka
Jan 2025	<u>Toward Resolving Heterogeneous Mixtures of Nanocarriers in Drug Delivery Systems</u> — ACS Nano
Aug 2021	<u>Systematic Review of Intrapartum Fetal Heart Rate Spectral Analysis</u> — Frontiers in Pediatrics

Full list of 14 articles and papers at m4rialoureiro.github.io

SPEAKING & PRESS

Speaking	Twelve-plus invited talks, panels and podcast appearances since 2021, including the Geek Girls Conference Portugal, Universidade de Aveiro, ENEEB XX, the Symposium on Bioengineering at FEUP, and <u>Prova Oral</u> on RTP Antena 3. Themes: AI in healthcare and the path from university into the field.
Press	Featured or interviewed in Expresso, Público, RTP Antena 3, RDP Internacional and Ordem dos Engenheiros, on AI in healthcare and representing Portugal at Davos 2026.

SKILLS, CERTIFICATIONS & LANGUAGES

Technical	Generative AI and agentic systems · large language models · machine learning · AWS cloud infrastructure · Python · end-to-end ML pipelines · biomedical and time-series data
Certifications	Claude Certified Architect – Foundations (Anthropic, 2026) · AWS Certified Machine Learning Engineer – Associate · AWS Certified AI Practitioner
Languages	Portuguese (native) · English (professional)